Explaining Concept Drifts in Business Processes: An LLM-Based Approach to Context-Aware Sense-Making in Process Mining

Jan Schaffner[1], Sandro Franzoi[1], and Jan vom Brocke[1]

University of Münster, Münster, Germany
{jan.schaffner, sandro.franzoi, jan.vom.brocke}@uni-muenster.de

Abstract. Concept drift detection in process mining identifies process changes but often fails to explain their underlying causes, creating a detection-interpretation gap. We address this problem by designing and evaluating the Concept Drift Explainer, a prototypical artifact that provides context-aware explanations for drifts. Using a Design Science Research methodology, we develop an LLM-based multi-agent system that systematically links drift signals from event logs to evidence from unstructured enterprise knowledge sources. We evaluated the artifact through a controlled experiment using real-world event logs and a synthetic context corpus. It achieved high retrieval accuracy, with Recall@2 = 100% and MRR = 0.875 across all evaluated drifts. Subsequent expert evaluation confirmed the artifact's practical utility, demonstrating its ability to substantially enhance analytical efficiency and explainability. In doing so, this paper contributes to research by operationalizing context-aware sense-making in process mining.

Keywords: Process Mining · Concept Drift · Large Language Models · Context · Explanation

Online Appendix

This appendix provides supplementary material for the main paper. It is structured as follows.

Appendix 1 contains a link to the CDE's project repository. Appendix 2 details the Design Science Research (DSR) process and evaluation design. Appendix 3 describes the empirical data and materials used in the evaluation. Appendix 4 documents the interview protocol and qualitative analysis procedure. Appendix 5 specifies the ten Design Objectives (DOs) that guided the artifact's construction. Appendix 6 reports the full results of the ex-ante evaluation rounds (Eval 1 and Eval 2). Appendix 7 reports the full results of the ex-post evaluation rounds (Eval 3 and Eval 4).

1 Project Repository

The source code, documentation on how to run CDE, and the event logs used for evaluation are available at the GitHub repository "concept_drift_explainer".

Table 1. DSR process phases, activities conducted, and outputs. Adapted from Peffers et al. (2007).

DSR Phase	Activities	Outputs
Problem Identification & Motivation	Structured literature review of concept drift detection and context-aware process mining; expert consultations to confirm practical relevance	Identification of the detection-interpretation gap
Objectives of a Solution	Derivation of ten Design Objectives from the literature; ex-ante expert validation via survey and semi-structured interviews (Eval 2)	Set of validated DOs (DO1–DO10)
Design & Development	Design of multi-agent LangGraph architecture; specification of the CDE information model (GraphState); integration of event logs, detector outputs, and synthetic context corpus; four iterative development sprints	CDE prototype artifact
Demonstration	Walk-through case study: incremental drift in the *International Declarations* sub-process explained by a compliance notice issued nine days prior to the drift onset	Illustrated end-to-end use case
Evaluation	Ex-ante (Eval 1, Eval 2) and ex-post (Eval 3, Eval 4) evaluation rounds; structured surveys and semi-structured interviews	Evidence of relevance, feasibility, technical performance, and practical utility
Communication	Documentation in conference paper and online appendix	Scholarly and practitioner communication

2 DSR Process and Evaluation Design

2.1 Research Methodology

This study adopts Design Science Research (DSR) as its overarching methodological framework, following the six-phase process model of Peffers et al. (2007). DSR was selected because it directly supports the dual objective of this work: generating prescriptive design knowledge through the construction of a purposeful artifact, while grounding that artifact in expert-validated problem relevance and practical utility (Hevner et al., 2004). The scope of this study encompasses a single, complete design-evaluation cycle, positioning the Concept Drift Explainer (CDE) as a proof-of-concept prototype; future iterations are expected to extend this cycle into broader empirical settings.

Table 1 maps each DSR phase to the concrete activities conducted in this study and their respective outputs. Figure 1 visualizes this process.

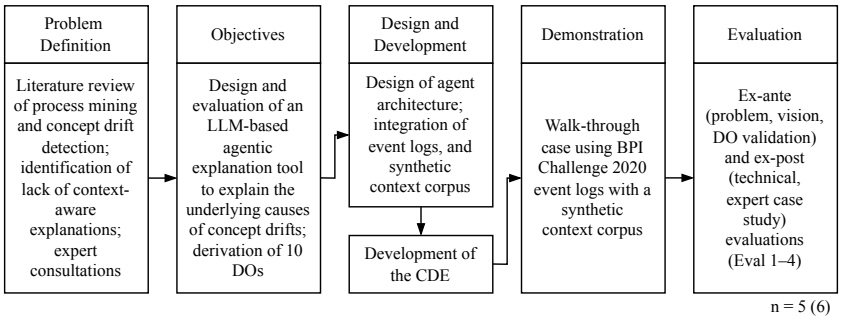


Fig. 1. DSR Process. Adapted from Peffers et al. (2007).

Table 2. Overview of all four evaluation rounds.

Round	Type	Focus	Participants	Methods
Eval 1	Ex-ante	Research problem & vision relevance	6 (P1–P3, R1–R3)	Likert survey, semi-structured interview
Eval 2	Ex-ante	Design Objective quality	6 (P1–P3, R1–R3)	Likert survey, semi-structured interview
Eval 3	Ex-post	Technical retrieval performance	— (automated)	Recall@k, MRR on 8 test cases
Eval 4	Ex-post	Practical utility & user experience	5 (P1–P3, R1–R2)	Case study, Likert survey, semi-structured interview

2.2 Evaluation Framework and Structure

Evaluation in DSR must address both the soundness of the design prior to implementation and its performance and utility after instantiation (Sonnenberg and vom Brocke, 2012). Following Sonnenberg and vom Brocke (2012), the evaluation was therefore structured as four sequential rounds (Eval 1–4), combining ex-ante and ex-post activities. Table 2 provides a structural overview of all four rounds.

Participant Profile. A total of six domain experts participated in Eval 1 and Eval 2 (P1–P3, R1–R3): three practitioners (two process managers and one process mining professional working with enterprise clients) and three researchers (two doctoral students and one postdoctoral researcher, all active in process mining or BPM research). Five of these six participants returned for Eval 4 (the postdoctoral researcher R3 did not participate in the final round). This balanced mix of academic and industrial expertise was deliberately sought to ensure that both the theoretical soundness and the practical applicability of the artifact were evaluated from complementary perspectives. Participant labels (P1–P3 for practitioners, R1–R3 for researchers) are used consistently throughout this appendix and the main paper.

Table 3. Statistical overview of the BPI Challenge 2020 event logs and detected concept drifts used in the evaluation.

Event Log (Subprocess)	Cases	Events	Drifts	Drift Types Detected
Domestic Declarations	10,500	56,437	1	Incremental
International Declarations	6,449	72,151	2	Incremental, Gradual
Prepaid Travel Costs	2,099	18,246	1	Incremental
Request for Payment	6,886	36,796	2	Incremental, Sudden
Travel Permit Data	7,065	86,581	2	Incremental, Gradual
Total	**33,000**	**270,211**	**8**	

Criteria for Ex-Ante Evaluation. Following Sonnenberg and vom Brocke (2012), the ex-ante rounds assessed five quality dimensions for both the research problem (Eval 1) and the Design Objectives (Eval 2): *accessibility* (can participants understand the artifact's intended scope without expert background?), *clarity* (is the formulation unambiguous?), *completeness* (does it cover all relevant aspects?), *feasibility* (can it be realized with available technology and data?), and *level of detail* (is the specification concrete enough to guide implementation?). For the research problem and vision (Eval 1), the criteria were additionally assessed along the dimensions of *applicability*, *suitability*, *importance*, and *feasibility*.

Criteria for Ex-Post Evaluation. Eval 4 assessed the artifact's practical utility across five dimensions, again following Sonnenberg and vom Brocke (2012): *applicability* (is the tool suited to realistic analyst workflows?), *effectiveness* (does it produce analytically useful output?), *efficiency* (does it reduce the time and effort required for root cause analysis?), *fidelity* (do outputs reflect the actual causal structure of the process?), and *overall impact* (does it change how analysts approach concept drift investigation?).

3 Empirical Data and Materials

3.1 Event Log Corpus

The evaluation is based on five real-world event logs from the BPI Challenge 2020 (van Donghen, 2020), documenting a travel expense claim process at Eindhoven University of Technology. This dataset was selected for three reasons. First, as a publicly available benchmark, it supports full reproducibility without requiring access to proprietary organizational systems. Second, the logs are directly processable by the CV4CDD-4D detector (Kraus and van der Aa, 2025), which provided the concept drift ground truth for the evaluation. Third, the non-trivial nature of the process—spanning multiple subprocesses, organizational units, and regulatory constraints—provides a degree of ecological validity.

Table 3 summarizes the statistical profile of each log and the concept drifts detected by CV4CDD-4D. Only drifts with a detector confidence score exceeding 50% were admitted as valid test cases, yielding eight distinct drifts across the five logs.

3.2 Synthetic Context Corpus

No public dataset exists that links event log concept drifts to their corresponding organizational documents. To enable a controlled technical evaluation (Eval 3), we therefore constructed a purpose-built synthetic context corpus of 35 documents simulating an organizational knowledge base.

Corpus Composition. The corpus contains eight *gold documents*, each engineered to serve as the sole correct explanatory document for exactly one of the eight detected drifts, and 27 *noise documents* designed as thematically related distractors. The noise documents cover topics adjacent to the gold documents (e.g., general travel policy, unrelated compliance updates, and organizational announcements from different time periods) to stress-test the system's filtering and ranking capabilities. The corpus was designed to cover two dimensions of the Process Mining Context Taxonomy (Franzoi et al., 2025): *organization-internal* context (e.g., restructuring announcements, leadership directives, process change memos) and *organization-external* context (e.g., regulatory notices, compliance mandates, external audit reports).

Generation Procedure. Documents were generated in a two-stage process. First, manual blueprints specifying the intended causal structure, drift type, and approximate publication date were authored for each of the eight gold documents. Gold documents were drafted with wording intentionally reflecting the associated drift type: for example, the gold document for the sudden drift contains emergency-register language, while the gold document for the incremental drift uses phased-implementation vocabulary. Second, a prompted LLM (GPT-4o) was instructed to act as a professional document author within the Eindhoven University of Technology administrative setting and generate realistic organizational communications consistent with the blueprints. Noise documents followed an analogous procedure, but were explicitly constrained to avoid wording that would make them causally relevant to any specific drift.

Document Formats. The corpus includes documents in three formats to test the system's multi-modal ingestion capabilities: .pdf (policy documents and regulatory notices), .docx (internal memos and announcements), and .pptx (organizational presentations, including slides with embedded organizational charts). Visual content in .pptx files was converted to text prior to indexing using a vision-enabled LLM (GPT-4o with image input) with a prompt engineered to extract business-relevant information from diagrams, tables, and charts.

Temporal Alignment. Each gold document was assigned a publication timestamp calibrated to fall within the CDE's skewed retrieval window of -14 to $+3$ days relative to the drift onset date. This asymmetric window reflects the empirical observation that organizational documents (e.g., policy announcements) are more commonly published before a process change takes effect than immediately after. Noise documents were assigned timestamps distributed across the full span of the event logs to ensure that temporal proximity alone could not serve as a reliable retrieval signal.

Ground Truth Labeling. Each gold document was tagged with a `gold_document` flag embedded directly in the evaluation metadata. This flag was used exclusively by the automated evaluation harness (Eval 3) to compute Recall@k and MRR, and was not accessible to any component of the CDE pipeline, ensuring unbiased assessment.

4 Interview Protocol and Qualitative Analysis

4.1 Interview Design and Conduct

All evaluation rounds involving human participants (Eval 1, 2, and 4) combined two complementary data collection instruments: a structured Likert-scale survey administered via Google Forms, followed by a semi-structured interview. This mixed-methods design was chosen to enable both quantitative comparison across participants and participants and in-depth qualitative exploration of the reasoning behind their ratings.

Recruitment and Scheduling. Participants were recruited via the professional and academic networks of the research team, with the aim of achieving a balanced mix of practitioners and researchers. Invitations specified the approximate time commitment (ca. 60 minutes per session) and the general topic (evaluation of a process mining prototype and the underlying research problem). Interviews were conducted individually and scheduled at the convenience of each participant. All sessions were conducted online via video conferencing.

Survey Instrument (Quantitative Component). Prior to each interview, participants completed a structured survey on a seven-point Likert scale. The survey for Eval 1 assessed the research problem and vision across the dimensions of *importance*, *suitability*, *applicability*, and *feasibility*. The survey for Eval 2 assessed each of the ten Design Objectives along the dimensions of *accessibility*, *clarity*, *completeness*, *feasibility*, and *level of detail*. The survey for Eval 4 assessed the artifact's practical utility along the dimensions described in Appendix 2, as well as additional items covering overall satisfaction, intention to use, trust in AI outputs (baseline and task-specific), and the perceived plausibility of the generated explanations.

Interview Guide (Qualitative Component). Each interview was guided by a semi-structured protocol developed in advance and adapted to the respective evaluation round. The protocol for Eval 1 and Eval 2 included opening questions on participants' professional background and experience with process mining, questions on each survey dimension (prompting participants to elaborate on their ratings), and closing questions on potential barriers to adoption and suggestions for improvement. The protocol for Eval 4 followed a think-aloud task-based structure:

1. **Context setting:** Brief introduction to the BPI Challenge 2020 event log and the known incremental drift in the International Declarations subprocess (approx. 5 minutes).

2. **Hands-on task:** Participants accessed the deployed CDE instance on their local machines and were asked to use the tool to formulate an explanatory hypothesis for the drift. No time limit was imposed; participants were encouraged to explore all available functions (timeline visualization, ranked cause list, source document links, export feature, and chatbot). The interviewer observed without intervening unless a technical issue prevented task completion (approx. 10 minutes).
3. **Debrief survey:** Participants completed the Eval 4 Likert survey (approx. 10 minutes).
4. **Semi-structured debrief interview:** The interviewer followed up on observations from the task, prompted elaboration on survey ratings, and explored hypothetical use cases and adoption barriers (approx. 20–30 minutes).

Recording and Transcription. All interviews were recorded with participants' explicit consent. Recordings were transcribed verbatim. German-language interview segments were initially translated using DeepL Translator and subsequently reviewed and corrected by the first author to preserve nuance and contextual accuracy.

4.2 Qualitative Analysis Procedure

The qualitative analysis of interview transcripts followed the thematic analysis procedure outlined by Braun and Clarke (2006). The procedure was applied inductively: rather than coding transcripts against a predefined framework, themes were allowed to emerge from the data. The analysis proceeded through the following phases:

Phase 1 — Familiarization. All interview transcripts were read in full multiple times. Initial observations and reactions were noted as free-form annotations to develop a holistic sense of the data before any systematic coding began.

Phase 2 — Initial Code Generation. Meaningful segments across all transcripts were identified and labeled with descriptive codes. Codes were generated at a semantic level (capturing the explicit content of statements) rather than a latent level. Example codes generated in this phase include: *efficiency gain through document triage*, *trust conditioned on source verification*, *confidence score opacity*, and *data provisioning as adoption barrier*.

Phase 3 — Theme Construction. Related codes were clustered into candidate themes. At this stage, candidate themes were assessed for internal coherence (do the codes within a theme form a meaningful pattern?) and external distinctiveness (are the themes sufficiently differentiated from one another?). Themes were iteratively merged, split, and renamed as this assessment proceeded.

Phase 4 — Theme Review and Refinement. Candidate themes were reviewed against the full set of coded extracts to verify that each theme was supported by sufficient evidence and that no important pattern had been absorbed into a broader theme prematurely. This phase also involved checking that the thematic structure was consistent with the quantitative survey data, using discrepancies between qualitative and quantitative findings as signals for further refinement.

Phase 5 — Naming and Definition. Each final theme was assigned a concise label and a working definition specifying what it captures and what it excludes. The four primary themes reported in the main paper are: (1) *Efficiency through automated document triage*; (2) *Trust constructed through source-level traceability (the Trust Paradox)*; (3) *Democratization of root cause analysis*; and (4) *Organizational preconditions and the Maturity Paradox*.

Phase 6 — Write-up and Integration. Themes were integrated with the quantitative survey results to produce the narrative synthesis presented in Section 5 of the main paper. Representative quotes were selected to illustrate each theme, prioritizing quotes that were both substantively informative and representative of the broader pattern observed across participants.

Quality and Reflexivity. No formal inter-rater reliability coefficient was computed, consistent with the exploratory and interpretive nature of thematic analysis in DSR (Braun and Clarke, 2006). To mitigate the risk of single-analyst bias, a reflexivity log was maintained throughout the analysis, documenting analytical decisions and the reasoning behind them. Findings were additionally triangulated against the quantitative survey data; cases where qualitative themes and Likert ratings pointed in the same direction were treated as convergent evidence, while divergences were examined more closely to understand their source.

5 Design Objectives

The ten Design Objectives (DO1–DO10) were derived from the literature review and subsequently validated in Eval 2. They are organized into three clusters: (I) Data and Context Layer, (II) Reasoning Quality, and (III) Trust and Transparency. Table 4 provides a concise specification of each objective, its motivation, and the primary agent(s) responsible for its fulfillment within the CDE architecture.

Table 4: Design Objectives (DO1–DO10): specification and architectural mapping.

DO	Title	Specification	Implemented via
Cluster I — Data & Context Layer			
DO1	Preprocessing of Process Mining Data	The system shall convert raw CV4CDD-4D detector outputs into plain-language, timestamp-enriched semantic abstractions suitable for LLM-based reasoning.	Drift Agent

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Table 4 – *Continued from previous page*

DO	Title	Specification	Implemented via
DO2	Contextual Data from Unstructured Sources	The system shall ingest and parse unstructured enterprise documents (internal and external context) into a vector index supporting semantic retrieval via RAG.	Context Retrieval Agent
Cluster II — Reasoning Quality			
DO3	Domain-Specific Knowledge	The system shall augment its retrieval with a curated BPM and process mining knowledge base to ensure that domain semantics inform the explanation.	Context Retrieval Agent
DO4	Multi-Source Reasoning Capability	The system shall aggregate, classify, and prioritize candidate causes from heterogeneous document sources using the Process Mining Context Taxonomy (Franzoi et al., 2025) and a re-ranking mechanism.	Context Retrieval Agent, Re-Ranker Agent, Context Mapper Agent
DO5	Agent-Based Task Decomposition	The explanation task shall be decomposed into specialized sub-tasks (drift parsing, retrieval, re-ranking, taxonomy mapping, synthesis) executed by dedicated agents with explicit state hand-offs.	LangGraph Orchestrator
DO6	Drift-Type-Specific Reasoning Logic	Explanations and reasoning prompts shall be tailored to the identified drift type (sudden, gradual, incremental, recurring) to avoid misinterpretation.	Explanation Agent
DO7	Handling Multiple and Overlapping Drifts	The system shall analyze and explain multiple co-occurring or successive drifts within a single event log, and identify cross-drift relationships.	Drift Linker Agent, Orchestrator, Frontend
Cluster III — Trust & Transparency			
DO8	Hallucination Warnings	The UI shall display a prominent, persistent warning that all AI-generated outputs are analytical hypotheses requiring human verification.	Frontend

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Table 4 – *Continued from previous page*

DO	Title	Specification	Implemented via
DO9	Explainability for Analysts	All insights shall be delivered in natural language; the system shall support interactive follow-up questioning via a context-grounded chatbot.	Explanation Agent, Chatbot Agent, Frontend
DO10	Output Traceability	Every generated explanation shall be anchored to a named source document with a direct link, enabling the analyst to audit and verify each claim against the original evidence.	Re-Ranker Agent, Explanation Agent

6 Full Results: Ex-Ante Evaluation (Eval 1 & 2)

6.1 Eval 1: Research Problem

The participants confirmed that the research addresses a highly relevant problem in process mining, centered on the critical gap between problem detection and causal explanation. Both researchers and practitioners highlighted that while current tools can identify process deviations, they fail to provide the context needed to understand the underlying causes or "triggers" (P2) This limitation was clearly articulated by P2, who pointed out the frustrating gap between identifying contributing factors and pinpointing the true root cause: "process mining can tell me that these are reasons, but what was the actual cause, the trigger [...]? That's what I'm missing." R1 confirmed that this lack of context is what prevents action, stating that when you "don't know why" a deviation is happening, "you can't make any improvements." This fundamental weakness of process mining, that it "uncovers problems, detects problems, but does not solve them" (R2), leaves analysts unable to implement sustainable improvements. This way, the "why" behind a deviation remains unknown (R1). P2 highlighted the significant business implications of this gap, particularly for clients investing heavily in process mining technologies. He shared a common customer frustration that directly questions the return on investment: "It is difficult, when we sell them the licenses for 50 to 100.000€, but then they say 'Okay, so I know there's something there, but what I don't know is how to fix the problem!'." However, one practitioner introduced a crucial nuance, arguing that the problem's relevance is context-dependent. In highly regulated environments with strictly formalized workflows, the relevance is considered "rather low" because unexpected process changes are less frequent. Conversely, in organizations where processes are described more superficially, the relevance is "significantly higher" (P3). P3 explained: "Well, for banks, I would say that because so much is already regulated [...] the relevance is rather low. The relevance for us [a large IT service provider], where the processes are described rather superficially [...] is significantly higher." This finding suggests that the challenge of explaining concept drifts is most acute in less rigid organizational structures. Furthermore, he added that "the larger an organization is, the more strictly processes are defined in

workflows and systems. [...] This means that these completely unexpected issues are no longer as relevant as they were 10 or 20 years ago." This suggests that the more structured the organization is, the less relevant the problem may become.

The problem was ultimately framed as a fundamental challenge of "sense making" within the broader field of Process Science (R3). The core need is to equip analysts with tools that allow them to more "easily and quickly" understand organizational changes by linking disparate pieces of information, a capability that is currently lacking (R3). This challenge is further amplified by the fact that explaining such process changes manually requires a rare combination of skills. As one participant noted, successful root cause analysis demands a "very high level of professional expertise in the processes, systems, and applications" (P2). This need for deep, specialized knowledge is coupled with the requirement for analysts to be broadly skilled across both technical and management domains. The difficulty of finding individuals with this hybrid profile is a known "major challenge" in the field, further deepening the problem for organizations (R3). R3 reported: "In the work [we did] it always came out strongly that this [finding individuals with broad knowledge] is, of course, one of the great challenges." The scarcity of such experts underscores the value of a tool that can augment analysis and lower the high barrier of entry to effective sense-making.

Although the theoretical relevance of the problem was confirmed, participants expressed significant concerns about the practical applicability of such a tool. P1 noted that "the approach only works if you have implemented process mining at a fairly high level of maturity." The tool was described as addressing an advanced problem, which one researcher characterized as "complaining at a high level" (R2), suggesting that for its benefits to be realized, "a lot of things have to be going right already" (R2). The consensus was that the proposed artifact is primarily relevant for a small segment of users who have achieved a high degree of maturity, while many companies are still at the beginning of their journey and "have not even defined their processes" (R2). This limitation manifests in two primary ways. First, the tool's target audience is users who engage in deep, analytical process improvement, but many organizations use process mining merely for superficial dashboarding. As one researcher noted, many users are only interested in seeing "how my process is right now and what the KPIs are on the dashboard. They are not interested in anything more" (R1). This means the tool would only appeal to those who use process mining "as it was intended [...] to sustainably improve processes" (R1). Second, the tool presupposes a high level of technical and data maturity that is often absent. Its approach "only works if you have implemented process mining at a fairly high level of maturity" (P1), yet many organizations "are simply not yet at the point" where their systems could reliably provide the necessary data (P1). One practitioner added that only "very few" customers are currently using process mining (P2). This reflects a well-known disconnect where "we in research are always very, very far ahead and deal with concepts that are not yet implementable in practice" (R3). Consequently, the tool is positioned as a solution for a future state of process mining adoption rather than a tool for the average organization today. Its application seems to remain in a "very theoretical area and not really in everyday business" (R2). Overall, participants confirmed the problem's relevance, though some voiced

reservations about the named reasons. Closing, P1 summarized this: "However, if process mining is truly mature in its application, then it is definitely a very relevant issue."

In summary, the qualitative feedback strongly affirms the relevance of the research problem, highlighting the "sense-making" gap as a significant practical and academic challenge that is amplified by the scarcity of human expertise. This confirmation was, however, qualified by two critical conditions identified by the participants: the problem's relevance depends on the specific organizational context and, more importantly, on a high level of process mining maturity that is currently rare in practice. To complement these nuanced qualitative insights and provide a quantitative measure of this assessment, participants also evaluated the problem's key dimensions on a seven-point Likert scale. The aggregated results of this survey are presented in Figure 2.

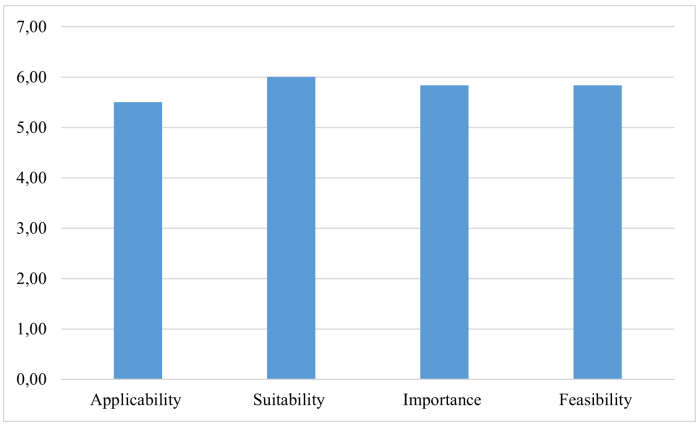


Fig. 2. Average participant ratings of the research problem across four dimensions on a seven-point Likert scale.

Participants rated the problem as highly relevant across all four dimensions. Suitability received the highest average score (6.0), followed closely by Importance and Feasibility (both 5.83). Applicability was also evaluated positively, though slightly lower (5.5). Overall, the results indicate strong support for the problem framing and confirm that participants perceived it as important, suitable, and feasible to address in practice. The lower score for applicability suggests that while the problem was considered relevant, some participants were more cautious about its immediate transfer into their organizational contexts.

6.2 Eval 1: Vision

The vision for the proposed artifact was met with strong, positive feedback from both practitioners and researchers, who regarded it as a pragmatic and valuable solution to the identified problem. The participants described the approach as "very practice-oriented" (P1) and a design that has immediately apparent benefits (R1).

A key value proposition highlighted was the tool's potential to quickly provide "an attempted explanation based on a large information base" for what might otherwise be a "cryptic hint" of a process change (P1). Furthermore, one researcher argued that such a tool could strategically enhance the standing of process mining in practice. By providing the missing "why", it helps to "refute the objection" that process mining only uncovers problems but doesn't help solve them, thereby making the approach itself "very relevant and interesting for practice" (R2). Beyond the explanation of past events, multiple practitioners independently identified an even greater potential for the tool to evolve into a predictive model. The participants envisioned a "super interesting scenario" in which the artifact could forecast future process outcomes based on contextual triggers. For instance, the artifact could state with a calculated probability that "because X happened, Y will happen to you tomorrow" (P2). This predictive capability was described as enormous, as it would offer the potential to not only anticipate issues but also to conclusively demonstrate the system-wide impact of specific changes (P2, P3). While the overall reception was enthusiastic, it was also grounded in a pragmatic view of the development process. While agreeing that the project is on a "super path", one researcher noted that its ultimate success will be determined by the "added value" it creates for analysts (R3). This frames the current vision as a highly promising "starting point and a first prototype" for a tool that addresses the fundamental need for users to "more easily and quickly understand changes in organizations" (R3).

Following the positive reception of the vision, participants moderated their optimism by identifying several significant challenges and limitations to the tool's practical feasibility. The most fundamental barrier that was raised was the pervasive issue of data quality and availability. One researcher emphasized this as the fundamental issue, citing the principle of "'Shit in, shit out'" and stating that without a "sound data basis, I cannot run any analyses" (R3). He then further elaborated, stating, "It's always the same thing: first you have to get the data right so that everything works. This situation is ubiquitous in every company" (R3). This problem is not only about the quality of existing data but also about the significant amount of contextual information that is never formally documented. As a practitioner noted, the tool's effectiveness is inherently limited when critical knowledge exists only informally "in the interpersonal space, in the air" and has never been formalized in a way that can be used as input for an LLM (P1). R2 described this issue by stating, "one is always dependent on this contextual knowledge, the knowledge that exists in people's minds." This informal context includes hard-to-capture human factors like "interpersonal tension" or the influence of a "team leader's personality" on process execution (P1). A researcher affirmed this, noting that much crucial "knowledge that is in people's heads" remains unrecorded, and the process to digitize it is a "long way off" (R2). The challenge of managing this data is further compounded by the increasing variety of relevant sources, including complex types such as "unstructured data, body data, [or] sensor data" (R3). Beyond the availability of raw data, a second major challenge lies in data integration. Even if contextual information is documented, it often resides in disparate, unconnected systems. A practitioner described the task of creating the necessary "mappings and connectors" to link information from various sources to a specific process problem as a "great challenge", noting that many companies have not undertaken such data map-

ping efforts (P3). Finally, concerns were raised regarding the real-world readiness of the solution. There is a significant gap between a functional prototype and a robust tool ready for enterprise deployment. As one researcher cautioned, while the development of a prototype is possible, the implementation of a solution that can be applied "one-to-one in practice" remains "difficult at first", and there is "still a lot to do" to bridge that gap (R2). These concerns serve to underscore a critical preliminary issue with the tool. This study operates under the assumption of the availability of all necessary data. However, it is acknowledged that this is a critical factor in a real business context.

To conclude the evaluation of the vision, participants identified two additional, user-centric challenges that would influence the practical adoption of the tool. First, the need for a careful cost-benefit analysis for each use case, and second, the overarching issue of building trust in its AI-generated outputs.

A practitioner raised the pragmatic concern that the tool's use requires economic justification on a case-by-case basis. P3 explained that before proceeding, a user must first ask, "how relevant are the deviations?" and then the "cost-benefit ratio" of a full analysis must be "carefully calculated" (P3). The second, more prominent challenge discussed was the need to establish user trust in the system's explanations. This sentiment was rooted in a general skepticism, with one researcher describing an LLM as a "big black box" and noting that the term "AI" is often "slapped onto everything" (R1). Consequently, the participants adopted a cautious "trust but verify" stance. While one might "in principle, trust it at first", they would "definitely double-check it" before drawing real-world consequences (R2). This need for human validation is especially critical when acting on the tool's suggestions, as one must be "very careful to check [...] if the answer is as accurate and correct as a human who researches for a longer time would find out" (P1). Ultimately, trust hinges on the "validity of the answer" (R3). While features like a confidence score were seen as a positive step, trust would depend on the tool's performance at scale with larger and more complex datasets (R3).

In summary, the qualitative feedback on the vision presents a dual perspective. The proposed artifact was met with significant enthusiasm and regarded as a highly suitable and valuable solution with future potential for predictive analytics. Conversely, this positive outlook was heavily moderated by a pragmatic assessment of the challenges ahead. The participants raised critical concerns, encompassing a broad spectrum of issues. These concerns included fundamental data-related barriers, such as quality, availability, and integration. They also encompassed crucial user-centric issues, such as the need for a clear cost-benefit analysis and the difficulty of establishing trust in AI-generated explanations. This is clearly reflected in the participants' quantitative ratings, which are presented in Figure 3.

Participants rated the vision positively across all four dimensions. Suitability achieved the highest average score (6.67), indicating strong agreement that the envisioned solution is congruent with the identified problem. Applicability (6.17) and Importance (5.83) also received high ratings, thereby underlining that the vision was seen as both relevant and meaningful for practice. The lowest score was observed for Feasibility (5.0), suggesting some reservations about the practical implementation of the vision in real-world settings. In summary, the results confirm that the vision was

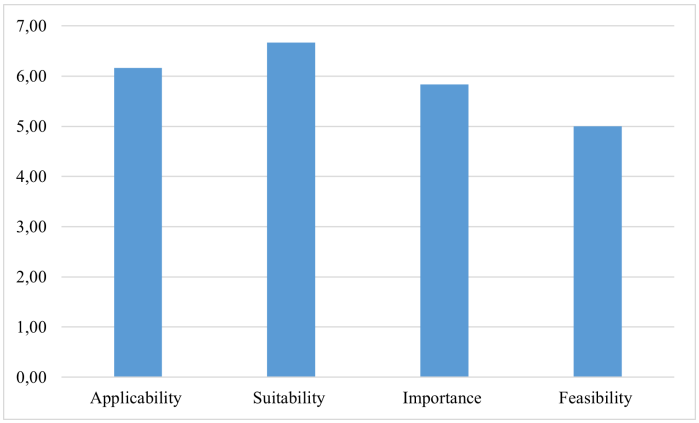


Fig. 3. Average participant ratings for the proposed vision on a seven-point Likert scale.

regarded as both suitable and valuable. However, the results also highlight concerns regarding the challenge of realizing it in practice.

6.3 Eval 2: Design Objectives

The proposed DOs were generally well-received by all participants, who consistently described them as logical, comprehensible, and well-aligned with the project's overall vision. R3 summarized this with the following: "They [the DOs] are all a good fit. Very understandable and sensible." However, the feedback received from the participants varied based on their respective backgrounds, revealing different areas of emphasis. Practitioners largely endorsed the objectives from a conceptual or "meta-level" perspective (P2). However, they also pointed out the need for more precise formulations in certain areas (P1, P2). A more significant concern was practical feasibility. P3 noted that, for complex interpretation tasks where real-world test data might be lacking, "it is very, very difficult to assess it [the system] in terms of usability and feasibility." The researchers evaluated the objectives as theoretically "feasible, good, and sensible" (R2), but underscored the importance of methodological rigor (R1). A strong consensus emerged among the researchers on the critical importance of the final cluster of objectives related to user interaction and trust. Features aimed at building user confidence, such as providing source citations and confidence scores, were unanimously highlighted as "extremely important" and "decisive" for the tool's practical adoption (R1, R2).

While the DOs were regarded as logical and reasonable on a "meta-level" (P2, R3), participants acknowledged that a deeper technical assessment of their granularity and feasibility was difficult without more expertise or real-world test data (P3, R1). The general consensus was that, in theory, the objectives appeared to be "achievable, good, and sensible" (R2).

The initial cluster of objectives related to data handling and the second cluster of the system's core analytical capabilities received feedback that was generally positive but highlighted areas needing clarification and pragmatic considerations. The

objective for Preprocessing of Process Mining Data (DO1) was seen as complete and feasible, thereby effectively preparing the data in a "well-readable format for the LLM" (P1). For Contextual Data from Unstructured Sources (DO2), a practitioner emphasized the importance of including details like timestamps (P1), while another raised a critical feasibility concern: "if you don't have real test data, what are you going to train it on?" (P3).

The feedback on the reasoning capabilities was mixed. The incorporation of Domain-Specific Knowledge (DO3) was briefly affirmed as "pretty important" (P1). The description for Multi-Source Reasoning Capability (DO4) was found to be unclear, with one participant feeling it sounded more like the "overarching goal" of the entire project rather than a specific objective (P1). This led to a resharpening of the DO, resulting in a significantly more precise objective. Conversely, the concept of Agent-Based Task Decomposition (DO5) was strongly supported, as breaking down a large, complex task into "smaller, sensible steps" was regarded as a "more promising" strategy (P1). The objective for Handling Multiple & Overlapping Drifts (DO7) was identified as a significant analytical challenge. Participants questioned whether the tool could distinguish between independent and interacting drifts, a detail that was unclear from the description (P1). This task was described as "even more difficult" as it would require disentangling overlapping causes and effects (P3).

The final cluster of DOs, which focused on the UI and transparency, received the most enthusiastic and unanimous praise. The objectives addressing the inherent uncertainty of AI outputs were seen as crucial for managing user expectations and building trust. Regarding Hallucination Warnings (DO8), one practitioner argued that since LLM outputs are "always potentially speculative", a warning should perhaps be the default and clearly express that all generated results may be "potentially incorrect" (P1). A researcher connected this objective directly to the critical theme of "Trust". The researcher stated that for a confidence score to be effective, it must be "understandable for the user where this confidence score even comes from" (R1). He highlighted the prevailing skepticism among users concerning solutions grounded in "AI logic", emphasizing the necessity to proactively build their confidence in the system (R1). The proposal for Explainability for Analysts (DO9), particularly through an interactive chatbot, was highly valued. Parti-cipants called the ability to ask follow-up questions "very good" (P1) and a "remarkable feature" that fosters trust (R1). P1 noted that "the opportunity to ask questions and have things explained in more detail, for example, 'Why are you bringing that up now?', is definitely very good." A researcher described his own workflow of interrogating chatbots to "grasp what it has done before I use anything", thereby underscoring the feature's practical utility (R1). Another researcher agreed that the combination of a confidence score and a chatbot would "significantly increase the user's trust in the tool" and make the solution "very practicable" (R2). Similarly, Output Traceability (DO10) was found to be of significant importance, with its ability to provide source links for its claims being described as "very important and clear" (P1). This feature was regarded as effective because it uses a familiar, trusted pattern from search engines, making the output easier for users to evaluate (P1). The overall sentiment for this user-facing cluster was positive, with one researcher summarizing it as "super important" from an application-oriented perspective (R1).

In summary, the qualitative evaluation confirmed strong conceptual support for the DOs. The feedback was most enthusiastic for the third cluster, where features designed to ensure transparency and build trust were unanimously praised as critical for the tool's adoption. In contrast, the feedback on the core data and reasoning objectives was more pragmatic, underscoring the necessity for enhanced clarity in certain descriptions and articulating valid concerns regarding practical feasibility. This pattern of high conceptual agreement, coupled with specific reservations about detail and implementation, is quantitatively reflected in the survey data presented in Figure 4.

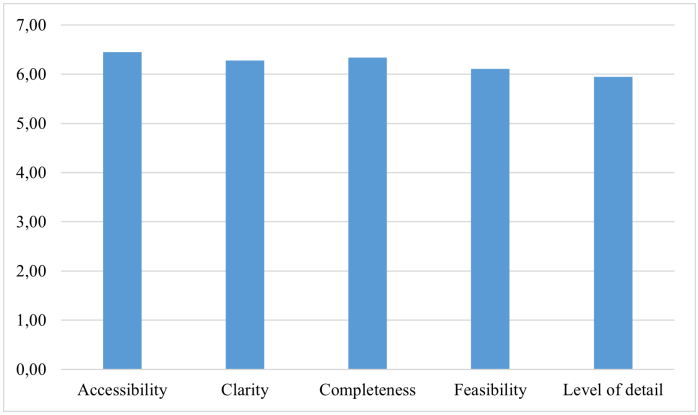


Fig. 4. Average participant ratings for the proposed Design Objectives on a seven-point Likert scale.

The objectives were evaluated positively across all dimensions, with accessibility (6.44), completeness (6.33), and clarity (6.28) receiving the highest ratings. This finding aligns with the qualitative feedback that the DOs were "logical, comprehensible, and well-aligned" with the vision. Feasibility received a strong but comparatively lower score (6.11), which reflects the practical concerns raised by participants regarding data availability and implementation. The lowest score for level of detail (5.94) quantitatively supports the recurring feedback that while the DOs were sensible on a "meta-level", more precision was needed in some areas. The survey results generally support the qualitative findings, indicating that the DOs are considered conceptually sound and well-conceived. The primary areas for attention identified by respondents pertain to the practical feasibility of the DOs and the granularity of their descriptions.

Further Considerations and Implications. Beyond the feedback on the specific evaluation objectives, participants raised several overarching points regarding the tool's responsible use, practical challenges, and its broader contribution to the field.

A strong consensus emerged that the tool should be treated as a hypothesis generator rather than an arbiter of truth. The risk of "apparent plausibility" was detailed from two perspectives (P1). P1 focused on the LLM's deceptive eloquence and noted that the problem arises when it "phrases things in such a way that they sound professional and

true, even though they are not necessarily so." R2, on the other hand, highlighted the danger of data-driven fallacies, where the tool identifies "apparent correlations" that seem logical based on the data, but a human would recognize as incorrect. To address this challenge, they proposed that the tool's output should be assigned a "suggestive status" and its formulation should be "diplomatically worded", employing phrases such as "it could be that..." (P1). The proposed workflow would be to use the tool to obtain "initial impulses" and "first ideas" (R2), which then serve as a "working hypothesis" for further human-led investigation (P1). As one researcher summarized, a user should always perform a "plausibility check" before drawing consequences, trusting the results only after they have been "cross-checked". R2 described, "I would always first take a really close look, perhaps quickly research it myself again, or do a plausibility check."

Participants identified several practical and ethical challenges. It was noted by one participant that the initial, human-led "pre-selection" of data to feed the system could introduce biases, which in turn could "influence the result" (P1). The most significant challenge raised was data privacy. This was described as the "first thing that immediately jumped out" as a concern (R2) and is a known difficulty with cloud-based LLM solutions (P1, P3). The practical challenges include gaining access to sensitive data, such as organizational charts and personnel files, and getting approval from IT security departments (P1, R2). Of particular concern was the ethical dimension, as participants warned against creating a system that might "spit out names" or unfairly blame individuals for process deviations (R2). Any output must be carefully managed so that an employee does not "walk into a trap just because the tool says 'XY was sick for two weeks, and that is the cause'" (R3). While these privacy issues were deemed less pressing for the initial proof of concept, they are critical for a real-world deployment (R2).

Ultimately, the project was regarded as a valuable contribution to its respective research domain. One researcher described it as a "good development for the entire process mining topic" because it "expands what has been done with process mining so far" (R2). This expansion is pivotal in addressing a common limitation where companies invest heavily in process mining tools but "in the end, maybe get a few dashboards out of it" (R2). By moving beyond description towards explanation, the tool aligns with the goal of "Creating Business Value" (R3), provided that ethical standards are met and the outputs are handled responsibly.

In summary, the participants recommended that the tool be used as a means to generate hypotheses to mitigate risks of potential false outputs. While critical challenges such as data privacy and user-induced bias were identified, the project was ultimately recognized as a valuable contribution to the field.

The insights gathered from the evaluation of the research problem, vision, and DOs (Eval 1 and 2) yielded valuable insights and provided clear guidance for the development of the artifact. The subsequent section now transitions from this conceptual validation to the practical implementation, presenting the results of Eval 3, which assesses the technical performance of the developed prototype.

7 Full Results: Ex-Post Evaluation (Eval 3 & 4)

7.1 Eval 3: Evaluation of Prototype System Metrics.

This section presents the third round of the artifact evaluation. This round constitutes an ex-post technical Eval 3 activity, as described by Sonnenberg and vom Brocke (2012). The objective of this stage is to ascertain the artifact's technical capabilities and performance in a controlled, artificial setting. This activity functions as a validation of the applicability, thereby confirming that the prototype instance is consistent with its DOs and successfully executes its core task before being assessed by end-users Sonnenberg and vom Brocke (2012). As demonstrated in the main paper, the prototype's implementation successfully fulfills all specified DOs. The subsequent evaluation, Eval 4, will then concentrate on the interplay between the artifact and its users in a realistic scenario to assess its practical utility.

The objective of this technical evaluation was to quantitatively assess the core performance of the CDE prototype in identifying the correct contextual cause for a given concept drift. The evaluation was conducted using a dataset based on five real-world event logs from the BPI Challenge 2020 (van Dongen 2020), which collectively contained eight distinct concept drifts. The system's knowledge base comprised a purpose-built synthetic corpus of 35 context documents, including eight "gold" documents designed to explain one drift each and 27 "noise" documents to simulate a realistic search environment. For each of the eight drifts, the CDE's task was to retrieve and rank documents from the knowledge base, with the objective of placing the single correct explanatory document at the highest possible rank. The performance of the system was measured using standard information retrieval metrics. First, Recall@k was employed to measure whether the correct document is present within the top k results. The MRR was utilized to evaluate the mean quality of the ranking across all test cases.

The results of the technical evaluation are summarized in Table 5. The CDE demonstrated a high degree of reliability and accuracy. A Recall@1 score of 75% indicates that in six out of eight cases, the system accurately identified and ranked the true trigger document as the single most probable cause. In addition, the Recall@2 score of 100% indicates that, across all eight test cases, the correct document was consistently positioned within the top two results. This is a crucial finding, as it demonstrates the system's capacity to significantly reduce the search space for a process analyst, thereby transforming a potentially arduous investigation into a swift validation of one or two candidate documents. The MRR of 0.875, near the ideal score of 1.0, substantiates the system's consistently high ranking quality, rather than being an artifact of a limited number of successful cases. This robust performance offers the initial quantitative evidence that the proposed artifact can effectively bridge the gap between detecting a concept drift and providing a context-aware explanation for the reason behind it.

Despite these promising results, this evaluation has several limitations. The primary constraint is the reliance on a synthetic corpus of context documents, as real-world ground-truth data linking event logs to organizational documents is not publicly available. Additionally, the anonymized nature of the event logs constrained the potential for deeper data integration, and the system lacked the capability to be trained or fine-tuned on authentic test data. Following the demonstration of the

Table 5. Concept Drift Explainer Prototype Performance Metrics ($n=8$).

Metric	Value	Description
Recall@1	75% (6/8)	The correct explanatory document was ranked in 1st place.
Recall@2	100% (8/8)	The correct explanatory document was ranked within the top 2 places.
MRR	0.875	The average reciprocal rank of the correct document across all test cases.

technical feasibility of the artifact, the next evaluation, Eval 4, will assess its practical utility and effectiveness in a realistic setting. This evaluation was conducted through expert interviews and a demonstration of the CDE with a case study.

7.2 Eval 4: Evaluation of the Final Prototype

The final evaluation, Eval 4, assessed the final prototype's practical utility, focusing on its applicability, effectiveness, efficiency, fidelity with real-world phenomena, and overall impact. In contrast to the preceding technical assessment, this final round followed the ex-post, naturalistic logic described by Sonnenberg and vom Brocke (2012), aiming to demonstrate the artifact's value and performance in a practical context.

To achieve this, the evaluation was conducted as a hands-on case study with five participants (3 practitioners, 2 researchers). The artifact was demonstrated through a realistic walk-through scenario centered on a known concept drift in a travel-expense log from the BPI Challenge 2020 event log set. Participants accessed the CDE via a deployed web application on their local machines and were tasked with using its functions to formulate a hypothesis explaining the identified concept drift. After completing the task, they provided feedback through a combination of brief structured surveys (n = 5; Likert 1–7 and multi-select items) and semi-structured interviews. The following analysis presents the findings, which have been thematically clustered and are supported by both qualitative insights and quantitative data.

Core Value Proposition: From Detection to Explanation

Overall Assessment. The initial and overall perception of the CDE prototype was exceptionally positive. Participants from both practice and academia immediately recognized its significant potential and relevance in addressing a critical, long-standing gap in process analysis. A practitioner summarized this sentiment by stating, "I rate the impact on process analysis as very high. The use of the tool definitely leads to a positive change" (P1). This enthusiastic reception was rooted in the tool's ability to directly address the core challenge that motivates much of the field. As one participant noted, the idea is "very good, because it addresses the central question that arises in both process controlling and process mining: What are the reasons for a change?"

(P2). Another researcher echoed this, affirming the tool's value in tackling the problem where analysts know what happened, but not why (R2).

Core Value Proposition. This core value proposition was seen as a potential "game changer" for the discipline (R1). The practical benefits were immediately apparent to the experts, who envisioned it as a "very useful tool" for process managers that would "enormously help to search for the causes of drifts and deviations more quickly" (R1). The feedback was not merely academic; it also sparked tangible business ideation. One participant described how the tool immediately started his "mind spinning", leading him to consider how it could be used as a unique selling proposition for his department, deployed with banking clients, and linked to existing applications (P2). Another, a process mining professional who works with many different processes, also saw "great potential" (P3). The participants' enthusiasm was consistently high, describing the prototype as "really cool" (P1, R2) and "super exciting" (P2). Survey results are consistent with this enthusiastic reception: overall satisfaction with the prototype averaged 6.2/7 (SD = 1.17), and the intention to use it was equally high at 6.2/7 (SD = 1.17). The implementation itself was considered impressive and functional, with one expert stating, "I found the tool very functional and easy to use. It's impressive what you have implemented there, and it also worked well" (P1). While acknowledging the current artifact is a first iteration, its value as a proof of concept was undisputed. It was praised as a "really good" proof of concept that is "super relevant for practice", is "definitely promising", and "has a lot of potential" (R2, P1, R1). The consensus was that the project is a "super topic" and that it is worthwhile to continue working on it in the future (P2, R2).

Impact on the Process Analysis Workflow. This positive reception reflects a shared expectation that a tool like the CDE would improve the analysis workflow by closing the "why" gap and moving from deviation detection to source-grounded explanation. Currently, process mining is often described as being at a level of: "'I'll show you that there's a deviation, but you have to think about the why yourself.' That's why this approach is really great." (P2). This gap frequently leads to a state of inaction, where analysis provides insights but does not translate into concrete improvements because the next steps are unclear (P1). The CDE helps to solve this problem by providing a crucial "filter" that is often missing from analyses, which otherwise become subject to subjective interpretations where the "actual reason is not recognized" (R1). By providing this missing context, the tool helps answer the critical question that follows many analyses: "'What do we do with this now? How do we proceed from here?'" (R1). In doing so, it has the potential to become a "real game changer" for the discipline (R1). While its value in simple cases is limited, its "real added value [...] becomes apparent when it comes to large, complex process structures in large companies", where it can help identify the multiple external influences on a process (R2). Overall, the anticipated "impact on process analysis is very high", and the use of the tool is expected to lead to a "definitively positive change" (P1). This sentiment was confirmed in the survey, where the perceived impact on process ana-lysis received a high rating of 5.8/7 (SD = 1.94). This quantitative result supports the impactful assessment from the interviews. While four out of five participants rated this impact as very high (6 or 7 out of 7), one participant rated it only as a 2 out of 7, taking a more cautious stance.

Core Analytical Applications and Use Cases. More specifically, this impact is delivered through the CDE's powerful support for the core analysis process, primarily by enhancing hypothesis generation and reducing complexity. The participants described the tool as "very well-suited for hypothesis formation, as it offers clear options for possible causes in the summary" (P1). This capability to enable "faster hypothesis validation" (P1) was identified as a significant advantage. By presenting a ranked list of potential causes, the artifact performs an "excellent complexity reduction", which helps analysts to focus their efforts (P1). The tool's true value was seen not in simple cases, where an analyst might already have a hunch, but in large, complex process structures. As one researcher noted, "The real added value of the tool becomes apparent when it comes to large, complex process structures in large companies. Then it is extremely useful to find out which three, four, or five external influences are changing a process" (R2).

This support manifests in key analytical applications that address common pain points. Participants noted the CDE is ideally suited for retrospective analysis (P3) in administrative processes that span multiple departments, where interdependencies are often unclear (R2), and for scenarios with "an extremely high number of parallel, co-occurring influencing factors" (P2). In such situations, where "many hypotheses are in the room" (P2), the CDE serves as a "direct springboard to initiate and implement improvements" by providing concrete starting points for investigation (R1). A key application identified was using the tool to validate explanations provided by business departments. As one practitioner described, the CDE can "jump into the breach" and help him "cross-check whether an argument like 'it takes so long because a billion new steps have been added' is really valid" (P3). This helps bridge the gap where analysts know how to improve a process technically, but are stuck asking, "What do we do now with this?" because the underlying "why" is missing (R1). When asked about specific applications, participants identified the primary use cases as root-cause search (100%), triage/early spotting (80%), monitoring/alerts (60%), and creating stakeholder-ready summaries (60%). Participants assessed the tool's fit as strongest for gradual (100%) and incremental (100%) drifts, and also high for sudden (80%) drifts, indicating its utility across a range of common change scenarios.

The User Journey: Efficient, but Trust-Reliant

Usability and Design. The prototype's usability and design were consistently praised for their simplicity and functionality. Participants viewed the straightforward interface not as a limitation, but as a significant strength. As one researcher noted, "The simpler, the better, because users make fewer mistakes" (R2). This sentiment was shared by a practitioner who found the tool "very functional and easy to use" (P1). The core workflow was described as clear and understandable. R2 described this as: "Ultimately, it's always the same: I select the right event log at the top, start the analysis, and then read through what the tool outputs. I think that's understandable." Specific design elements like the "summary and visualization" (P1) and the concise "Executive Summary" sentence (R1) received positive feedback. Quantitatively, problem-solving support scored high: participants strongly agreed that the tool was effective at focusing their attention (6.2/7, SD = 0.98) and reducing the problem's complexity (6.0/7, SD =

1.10). The explanations provided in the case study were described as "plausible" (P2), a finding supported by a high average plausibility rating of 6.6/7 (SD = 0.49). However, there was also specific criticism regarding the presentation of evidence snippets, which one participant found to be "a bit chaotic" due to the unformatted block of text (R1).

Accelerating Root Cause Analysis. Besides the usability and design, another central theme that emerged from the evaluation was the prototype's profound impact on analytical efficiency and the potential for significant time savings. Participants unanimously agreed that the CDE would drastically reduce the manual effort currently required for root cause analysis. The core benefit lies in its ability to automate what is now an "extremely difficult and time-consuming" process of manually sifting through documents to establish connections (P2). One practitioner described the current state as a hunt where "you have to search through individual documents, find the right places, and in the end can only form hypotheses" (P2). The CDE directly addresses this pain point, with one researcher stating, "the manual research effort is definitely reduced. Instead of looking at 30 documents, I can concentrate on the one or two most important ones and save a lot of time" (R2). This was described as an "incredible corner cutter - a huge advantage" (R1). The entire workflow was perceived as fast and seamless. As one participant noted, features like "the direct links to the context documents, the ranking of the results, and the highlighting of the quotes save a lot of time. The whole process felt quick and fluid, and nothing slowed me down" (P1). When asked to identify the most significant time-saving UI elements, participants selected the links to context documents (100%), the ranked list of causes (80%), and the timeline view (60%). The tool was seen as enabling a much faster validation of hypotheses and a more rapid search for root causes (P1). However, this efficiency gain was also presented with important caveats. One expert argued that the time saved is conditional on the organization's structure. If the process manager and designer are the same person, this expert "usually knows the reasons for his concept drift himself" and may be faster using their own knowledge (P3). In larger organizations, where the process manager and designer are different people, the tool becomes "very useful" for bridging knowledge gaps and thus avoiding the cascade of "meetings, queries, follow-up questions, and so on" (P3). Therefore, the tool's greatest utility emerges in larger organizations where roles are separated and knowledge is siloed. While it is questionable whether employees without any process management responsibilities would proactively conduct such analyses, the CDE is positioned as a vital instrument for the primary target group of process analysts who are tasked with investigating processes outside their immediate domain of expertise (P1). A more significant concern was raised that the efficiency benefit could be nullified if the tool's suggestions are not accurate. If an analyst has to manually validate every document because the tool's relevance matching is poor, it would "contradict the purpose of the tool, which is to make me faster" (P3). This underscores that the promised efficiency is directly tied to the system's reliability, a point acknowledged even by the most optimistic participants who would still conduct their own validation (R1, R2).

Effectiveness of the Results Presentation. While the tool's efficiency was praised, the effectiveness of its results presentation elicited more complex and nuanced feedback. The tool was very effective at focusing the user's attention on the most critical

information. Participants reported that the system "quickly drew my attention to the most likely cause, especially through the ranking and the confidence numbers, like the 99%" (P1). The confidence level served as an "important impulse that directs the view to the most likely reasons" (P2). The most valuable function was the system's ability to generate a testable hypothesis based on its sources, a capability that received a perfect 7.0/7 score (SD = 0.00) from participants. This analytical support was seen as a way to democratize root cause analysis and augment analyst capabilities, regardless of their level of domain expertise. Participants highlighted that the tool can "compensate for skill deficits in employees who may lack the intuition or deeper knowledge for such analyses", making it a "very, very good supplement" to the existing analytical toolkit (P2). The participants demonstrated this by successfully finding plausible explanations during the case study. As one practitioner noted, this was achieved even "without deep expertise in travel expense accounting", and it was concluded that the tool is "particularly helpful for analysts who lack domain knowledge" (P1). This was seen as especially valuable for knowledge retention in organizations where experienced employees are retiring and taking their expertise with them (P1). A practitioner emphasized that this is precisely what is missing in the market today: "These [existing] tools often just show you that something happened - period. [...] The great advantage here is that a possible causal relationship is shown" (P2). This was reinforced by a researcher who noted that the tool's explicit framing of its output as a "hypothesis" rather than a causal guarantee effectively "defuses a potential point of attack" (R1). The quality of the explanations was rated highly for completeness (6.0/7, SD = 1.26) and consistency with data sources (6.8/7, SD = 0.40). However, participants also identified significant weaknesses. The most critical functional gap was the lack of information about the concept drift itself. One expert stated, "I was missing what the change actually includes. For example, were new activities added or was the description changed? To be able to perform a target-actual comparison, this information is missing" (P3). Crucially, despite this missing context, participants still felt the tool enabled them to name a concrete change likely triggering the drift (6.2/7, SD = 0.84).

Trust. The presentation of results naturally led the discussion to the crucial and complex issue of trust in the artifact's AI-generated explanations. Survey data revealed a significant increase in trust through the use of the CDE: while baseline trust in explanatory AI was mediocre at 4.0/7 (SD = 1.26), task-specific trust in the CDE was substantially higher at 5.6/7 (SD = 1.36). This fits with the consensus among participants who took a pragmatic stance of "trust but verify". While they saw immense value in the tool's output, they uniformly stated that they would not blindly accept its conclusions. A researcher summarized this workflow: "I would still do my own research afterward, but for a first overview, I would definitely use it" (R2). This sentiment was echoed by a practitioner who, even if a result had 100% confidence, would "still check it manually", admitting that his "trust in AI is not yet at a point where I would trust it blindly" (P2). Notably, while no parti-cipant reported experiencing a hallucination (0/5), all still emphasized the need to verify explanations via source links. The immense value of the CDE, therefore, was not seen as providing an infallible final answer, but rather as a powerful tool for enhancing analytical efficiency. The "enormous advantage is that you are directly pointed to the

three or four most likely reasons" (P2). This ability to accelerate the initial phase of investigation was highly valued. Participants confirmed that the explanations provided were professionally sound and plausible when checked against their own expertise, as P1 stated: "Of course, you should check the results using your own expertise, but they have always seemed plausible so far." Crucially, the evaluation revealed that the single most important feature for building user trust was output traceability. The ability to link back to the source documents was universally praised as the mechanism that makes the system's claims verifiable and transparent. One practitioner explicitly stated that while their general rating for the tool was a 5 out of 7, their "trust in the results is a 7, because you always have the sources and can form your own picture" (P1). This was strongly corroborated by a researcher who found the document attachment to be "very important", adding that "without this document, it would be difficult to trust the information" (R2). R2 concluded, "I think that's a very important point of the whole system: you can go directly to the original source. [...] This ensures transparency and builds trust in the application" (R2). Participants recognized that the challenge of trust is not unique to this artifact but is a pervasive issue for nearly all AI tools (R1). One researcher noted that many AI applications suffer from this "black box problem", and achieving full transparency of an AI's "thought process" is likely not feasible (R2). Another participant articulated a trust paradox: a decision-maker would likely trust a process manager who came to the same conclusion after a lengthy manual analysis more than one who used the "shortcut" of an AI system, even if the result was identical (R1). This "residual doubt - especially among decision-makers" (R1) necessitates that the final analytical judgment remains with a human. The consensus was that the tool provides suggestions, but the "ultimate analysis should still be approved by me as a human" (R1). The analyst must act as the "human-in-the-loop" who makes the final decision, because there is no "compelling logic" for why a decision was made (P3).

Critique of Confidence Scores. Regarding the topic of trust, a major critique targeted the confidence level. The confidence level shown for each cause was described as a "black box" that made it "extremely attackable" (R1). Without a simplified explanation of how the score is calculated or a scale for interpretation ("When is a confidence level considered good and when is it considered bad [...]?"), it was perceived as a "metric that I don't understand" (R1). The accessibility of the original source documents was seen as the key feature that mitigated some of these issues. This is strongly supported by the survey, where source links (100%) and the ranking of causes (80%) were cited as the primary drivers of trust, while the raw confidence percentage was seen as hard to interpret, which reinforces the need for a more explainable, banded scale. When the AI-generated summaries were occasionally difficult to understand due to jargon, the "linked original documents were very clear and understandable" (P1).

Overall Effectiveness and Intention to Use. Ultimately, despite the critiques regarding the transparency of certain features, the overall effectiveness of the tool was strongly confirmed. One researcher noted that while she had to first familiarize herself with the context, an internal employee who already knows the documents could "quickly scan over them", making the application "even easier" (R2). She concluded, "Even then, I was able to answer the question in the end. This means the explanation

itself was definitely good" (R2). This fundamental effectiveness, combined with the significant efficiency gains, resulted in a near-unanimous and enthusiastic consensus among all participants that they would use the CDE in their professional work. The intention to use the tool was exceptionally high, with one practitioner stating, "To give an assessment of the intention to use, I would definitely use the tool for a drift investigation. I would rate it a 7 [out of 7]" (P1). Quantitatively, participants anticipated that the tool's primary organizational impact would be as a shortcut for root-cause searches (100%), leading to higher-quality analysis (80%) and direct work savings (80%). The primary driver behind this strong intent was the tool's ability to dramatically increase the efficiency of what is currently a highly manual and frustrating process. As one participant explained, "Yes, I would definitely use it. The main reason is that it is extremely difficult and time-consuming to establish the connection in such drifts. [...] I believe the tool can significantly help in saving work time here" (P2). The CDE was seen as a powerful way to get "quick and uncomplicated helpful hints" (P1). This aligns with the overall view of the CDE as a powerful assistant that augments, rather than replaces, the human analyst. Participants confirmed that the ideal approach is to use the CDE as "a tool that reduces the effort for manual research", while acknowledging that "at the end of the process, a human is always needed", especially to avoid spurious correlations (R2). This was augmented by the survey results, which show that the tool's output was seen as highly actionable: given an explanation, 60% of participants would act on it with only minor additions, while the other 40% would trigger further validation. Crucially, none chose to simply file the information away without action. Furthermore, the tool's utility was also seen as context-dependent, particularly in relation to organizational structure. One expert noted that its value increases in larger companies (from 300-400 employees) where the roles of process manager and process designer are separate. In such settings, "such a tool would probably be - depending on the cost, of course - very useful" for bridging the knowledge and communication gap between these distinct roles (P3).

Path to Adoption: Hurdles, Economics, Roadmap, Vision

"Must-Have" or a "Nice-to-Have"? The discussion on the tool's necessity sparked a nuanced debate, revealing that its perceived value is highly dependent on the organizational context. The central theme that emerged was that the artifact's classification as a "nice-to-have" or a "must-have" is directly linked to an organization's process maturity. A practitioner articulated this condition clearly: "That depends entirely on the process maturity of the organization. From the perspective of a dedicated process manager, the tool is a necessity. However, if the organization as a whole does not recognize the need for process optimization, it will be seen as a pure 'nice-to-have'" (P2). A researcher concurred, stating that the classification "depends on how well a company's processes are already structured" (R2). One strong argument was that the tool is becoming increasingly essential due to growing organizational complexity. As one participant explained, "Especially because the number of possible causes for a drift is always increasing, you quickly lose track. [...] That's why, in my opinion, such a tool becomes increasingly important and necessary", particularly in large companies

where an individual cannot possibly keep track of all concurrent events (P1). In such environments, a central tool that collects and analyzes all data is "extremely helpful" (P1). For this reason, some participants leaned firmly towards classifying it as a necessity, or at least a highly valuable asset that would "enormously help to search for the causes of drifts and deviations more quickly" (R1). However, this view was not unanimous. One process mining professional offered a more critical perspective: "Even if it sounds harsh, for me, it is a tool among many and belongs more in the 'nice-to-have' category" (P3). He expressed concern that companies mature enough to benefit from such a tool would have quality standards that might be difficult to meet and that he "would have a hard time recommending a permanent license fee for it" (P3). This divergence revealed an interesting paradox in the tool's perceived value. One researcher suggested that "the more opaque your process structure is, the more necessary such a tool becomes to find out why a process is not running" (R2). Conversely, another participant argued that only in a truly process-oriented organization, where deviations from set targets are actively investigated, does "such a tool become a necessity" (P2). This creates a challenging dynamic: the organizations that might need the tool most (those with less structured processes) may not have the maturity to recognize its value, while those with the maturity to value it may have more structured processes to begin with. Despite this, the business potential was strongly affirmed by one practitioner, who saw "enormous value" in leveraging the tool as a "huge competitive advantage" for his own company and expressed a clear interest in a future collaboration: "Keep us in the loop, this could be a potential use case for us" (P2).

Prerequisites and Limitations for Real-World Adoption. However, before this business potential can be realized, participants engaged in a critical discussion about the real-world prerequisites, hurdles, and limitations that stand in the way of adoption beyond the fundamental question of cultural readiness. Survey data quantitatively confirmed the most critical non-negotiables: participants unanimously selected log data coverage (100%), context data coverage (100%), and clarified data privacy (100%) as essential preconditions for adoption. The most significant practical hurdle identified in the interviews is the practical challenge of providing and maintaining high-quality data. This was seen as a critical external factor influencing the tool's reliability. The core issue raised was not a flaw in the tool itself, but its dependency on the quality and temporal accuracy of the organizational knowledge it ingests. This concern was framed as a fundamental question of data integrity: "Who guarantees that the collected data is even correct? One must somehow verify that the results are not completely absurd" (R1). A practitioner provided a concrete and common real-world example of this challenge: retroactive documentation. He explained that in practice, "documents like work instructions are often changed retroactively, as operational implementation is usually more important than immediate documentation" (P3). This creates a significant problem for the CDE's analysis, as the timestamp on a document may not reflect the actual time a change was implemented. As the participant noted, "This creates an inaccuracy for your analysis, as it is logically impossible to find a cause that lies in the future. This is therefore a pure documentation issue" (P3). Such scenarios underscore the necessity for the "human-in-the-loop" to apply additional contextual knowledge and identify potentially non-logical findings (P1, P3). Another significant prerequisite and practical

hurdle is the immense effort required to provide and maintain the necessary context data. Participants noted that for the CDE to function, "all relevant documents must be made available in advance" (P1). In a real-world scenario, this knowledge base would not contain 35 documents, but rather "35.000" (P2), spanning a wide variety of formats and authors (P2). The core challenge is that most companies do not have this information readily available. As one expert described, "If you ask a company if it has all context documents centrally in one database, the answer will usually be 'No.' The most important documents are here, others are there, and the rest comes via email" (P3). The effort to retroactively feed and continuously maintain this data was seen as a "huge challenge" and a significant additional cost, leading to the conclusion that "the effort is simply very, very high for the benefit" (P3). These data challenges are a key factor contributing to the currently low market penetration for this type of advanced analysis, as already outlined in Eval 1 and Eval 2. Several participants confirmed that many companies are simply "not that far along yet" (P1), with some still lacking even basic digital record-keeping (P3). The tool's value is therefore highly dependent on an organization's process maturity. As one practitioner explained, "If an organization is intensively engaged with process management, it will consider this tool a necessity. If the attitude is 'We just have processes and now another tool,' the necessity will not be recognized" (P2). Beyond data availability and market maturity, data privacy was identified as another "huge hurdle", particularly in an EU or German context (P3). The need to upload "large and important amounts of data into a cloud, which is often located outside the EU", presents significant practical and legal challenges (P3). This is compounded by questions regarding data storage duration, processing rights, and the indirect provision of data for training purposes (P3). Alongside data privacy, other foundational prerequisites such as performance and the management of access rights must be addressed before any operational deployment (P1).

Challenge of Economic Justification. Despite the clear analytical benefits, participants raised another significant real-world challenge: articulating a compelling economic argument for the tool's adoption. One expert highlighted the core difficulty: "It is incredibly hard to argue that a tool like this saves more costs but doesn't earn more money. These savings are not clearly quantifiable, which makes the tool unattractive for business" (P3). This reflects the practical reality that process improvement is not a purely technical exercise but also involves "internal politics, costs, and persuasion" (R1). This feedback underscores that for the tool to succeed beyond a proof of concept, a clear business case must be developed that can effectively communicate the value of its efficiency gains to decision-makers. However, participants suggested that the CDE itself provides the tools to build this very business case through its potential strategic and communicative capabilities.

Beyond Analysis. In addition to its analytical power, participants saw significant potential in using the CDE as a strategic tool for demonstration, justification, and communication. It can serve as a powerful "demonstration instrument" to vividly showcase the value of process mining to colleagues or management who are less familiar with the topic (P1). By presenting "concrete events from the organization" instead of abstract numbers, it makes the link between internal actions and their

process impacts visible (P1). This capability can even be used to justify investments in foundational but often perceived as "boring" topics like data management, by showing they are the prerequisite for such powerful analyses (P1). This communicative function extends directly into change management. Participants suggested using the prototype as a "Wiki or explanation tool for users" to proactively address the "emotional pain" that often accompanies a process drift (P3). For instance, when a change is mandated by an external regulator, the tool's neutral presentation can depersonalize the issue and foster acceptance more efficiently than repeated training sessions (P3). This neutrality of the chatbot is a key advantage, as users can seek clarification without having to admit they haven't read a manual (P3). In this way, the CDE becomes a powerful ally in navigating the very organizational hurdles it faces.

Functional Enhancements. Acknowledging all these significant real-world challenges, the discussion then turned to a constructive and forward-looking roadmap for the artifact's future. The final part of the evaluation focused on gathering critical feedback and concrete suggestions for the future development of the artifact. The most frequently cited suggestion was for the tool to evolve beyond explanation and towards actionable recommendations: "A feature that makes suggestions for the next steps would be very helpful." (P1). Participants from both practice and research expressed a strong desire for a feature that could generate "concrete action proposals" (P1). They argued that knowing the cause of a problem has not yet solved it, and a "direct feature that suggests possible next steps would be a useful addition" (P1). This would allow the tool to "directly enter into practical process optimization" by not just identifying causes but also making "improvement suggestions" (R1). Participants also proposed enhancements to deepen the tool's analytical capabilities. A key suggestion was to measure the success of an implemented change (intended changes), not just the causes of unintended changes. As one expert argued, the value would be much greater if the tool could answer such questions in this context: "Did the change actually take hold? Did I become compliant again, or did people revert to the old process?" (P3). He noted that finding out that "3 months later, 95 percent of the employees are still doing it the old way" is often a "much more important piece of information" than just knowing the root cause of a concept drift (P3). Another suggestion included incorporating data on process participants to gain another analytical dimension. One practitioner framed the need for a feature like this as: "Are more or fewer people involved in the process? [...] It hurts less if the process takes 3 to 4 days longer, but it is much worse if a person has to invest another five minutes or pass the task on to three other people [...]." Further suggestions added the ability to recognize and combine multiple causes (instead of just one trigger document per concept drift, as currently implemented in the proof of concept) (R2), and graphically integrating the process model into the results page for easier comparison (P1). Finally, the discussion culminated in a long-term vision for the CDE as a core component of a fully autonomous process optimization system. One researcher described the "dream for every process manager": an "agent system" that sits on top of the process mining data and "handles virtually the entire workload" (R1). This envisioned system would perform the analysis, detect deviations, find their causes, and output recommendations. In a

fully digitized process, the system "could perhaps even fix the deviation itself", with the "human-in-the-loop" only needing to approve the action with a single click (R1).

UI/UX Enhancements. In addition to functional enhancements, participants provided several concrete suggestions for optimizing the UI to improve the clarity and interpretability of the results. The feedback centered on creating a better visual hierarchy and, most critically, on contextualizing the confidence score. To better focus the user's attention, it was suggested that low-probability results (e.g., below 50%) could be "optically highlighted", for instance by displaying them in a progressively lighter shade of gray or a smaller font (P1). The most detailed feedback concerned the confidence level, which, without context, was described as a "metric that I don't understand" (R1). A practitioner strongly argued for a "simplified, understandable explanation" of how the score is generated to make the tool less of a black box (P3). To help users interpret the values, he suggested adopting familiar visual systems like "traffic light colors" to classify whether a score is good or bad, asking, "I know 99% is probably good [...]. But what about 44%? No idea, is that good? It could also be bad" (R1). This need for better classification of the confidence level would determine how closely a user would scrutinize the results and could even be implemented as a filter (R1).

Marketability. Thus, while the path to a fully mature product involves overcoming these hurdles and implementing these suggestions, the participants' final assessment of the prototype's market potential was decidedly optimistic. Despite these significant hurdles, participants saw a clear path to marketability. The true limits of the tool can only be assessed in a "real, very complex use case" (P1), and the prototype serves as a powerful proof of concept that this direction is viable (R2). If the considerable challenges of data provisioning and privacy can be solved, the artifact is considered a "sellable solution if it is scaled up" (R1). For an organization that can provide the necessary data access, the tool offers "enormous value" and could become a "huge competitive advantage" (P2), prompting one practitioner to express concrete business interest: "Keep us in the loop, this could be a potential use case for us" (P2).

8 Summary of Evaluation Activities

The multi-stage evaluation of the artifact was designed to ensure rigor and relevance. It progressively validated the CDE from its conceptual foundations to its practical application in a realistic setting. The four evaluation rounds followed a logical progression in line with DSR principles, moving from ex-ante conceptual validation to ex-post technical and practical assessments Sonnenberg and vom Brocke (2012).

The process began with conceptual validation in Eval 1 and 2. Through expert interviews and surveys, these initial ex-ante rounds confirmed the high relevance of the research problem and the conceptual soundness of the proposed vision and DOs. While the experts affirmed that a tool addressing this gap would be highly valuable, they also moderated this positive outlook by identifying critical real-world dependencies. These include the prerequisite of high organizational process maturity and the significant challenge of data availability and quality.

Following this, Eval 3 provided the technical validation of the prototype. In this ex-post evaluation, which was conducted in a controlled environment, the artifact demonstrated strong technical feasibility and high reliability. This round successfully proved that the designed solution could be built and accurately identify the correct contextual causes of concept drifts from a knowledge base by achieving a Recall@2 score of 100% and a high Mean Reciprocal Rank (0.875).

Finally, Eval 4 provided crucial practical validation through a hands-on case study with experts. This ex-post, naturalistic evaluation confirmed the prototype's "game-changer" potential and its utility in real-world workflows. The findings showed that the tool's value comes from an efficient "human-in-the-loop" model where trust is based on source-based transparency rather than blind trust in AI. However, the final round also revealed fundamental hurdles to adoption. The most notable hurdles were the immense practical effort of data provisioning, the strategic challenge of data privacy, and the recurring theme of organizational maturity. While the tool's core function was deemed highly effective, the lack of transparency regarding the "confidence level" was identified as the most significant critique of the current implementation.

Overall, the evaluation activities successfully validated the artifact at each stage of its development, confirming its value and identifying a clear path for improvement in the future. A detailed synthesis and discussion of these findings and their broader implications will be presented in the following chapter.